Fred Hutch Cancer Center

*Translational Data Deep Dive*

# Building Computational Workflows

March 11th, 2026, DaSL Lounge and Teams

# What is a Deep Dive?

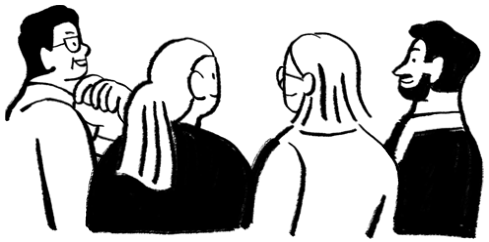
A hands-on, interactive quarterly series to help build practical data science skills at Fred Hutch

## Learn Concepts

*Understand the **what**, **why**, and **when** behind data science approaches.*

## See the Tools in Action

***Live demonstrations** using Hutch-supported tools and infrastructure.*



## Ask & Engage

*These sessions are interactive  ask questions in person or online using Slido.*

## Get your hands dirty!

*Try things yourself during the co-working time*

# Who we are?

*And why should we be teaching you about computational workflows?*

**Sita:** I biologist by training, I have used WDL's for 5 years for analysis of bulk genomic and single-cell data.

**Taylor Firman:** A bioinformatics software engineer that has used Bash/SLURM/Python in biophysics & genomics, but is much happier having discovered WDL and Docker.

**Emma Bishop:** A bioinformatician who learned workflows to run single-cell pipelines in the cloud and now uses WDL instead of bash scripts to analyze T cells.

# Who else is here to help today

**Scientific Computing**

- Dan Tenenbaum, Scientific Computing
- Michael Gutteridge, Scientific Computing

**PROOF Workbench**

- Scott Chamberlain, OCDO
- Sean Kross, OCDO

**Training Team**

- Chris Lo, OCDO
- Ted Laderas, OCDO

**Cirro**

- Sam Minot, Cirro

# Before we begin…

*Some housekeeping to make this session work for everyone*

- You can find the materials associated with this Deep Dive @
  - Deep Dive Pages
  - GitHub

- We will be using Slido
  - Just go to slido.com and enter this code #8758973
  - More details on how to use Slido here

- We follow the Training Teams DaSL Participation guidelines

# Learning Objectives

## Recognize When Workflows Are Useful

*Learn **when** a computational workflow is the right solution for your analysis*

## Don't Build Workflows from Scratch

*Know about how you can leverage resources like the **WILDS WDL Library** to assemble your own workflow*

## Understand the Structure of a WDL

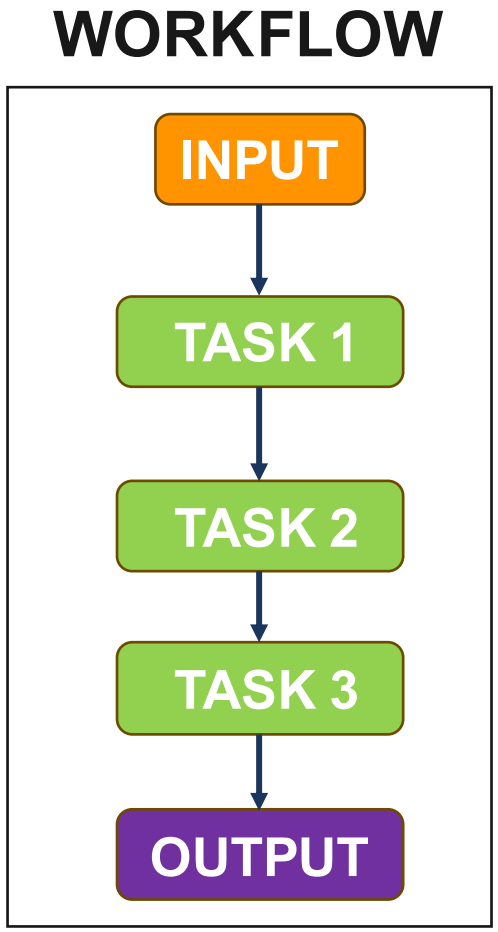
*Learn how to read and interpret the key components of **WDL**, a language to write workflows in*

## Learn how to run workflows at Fred Hutch

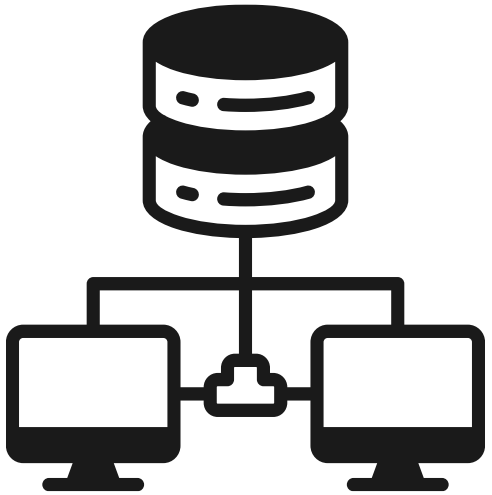
*See how workflows can be run using **PROOF Workbench** and Hutch infrastructure*

# What is a Workflow?

A workflow describes all the steps needed to turn input data into results.



# What are the advantages of writing and running a workflow?

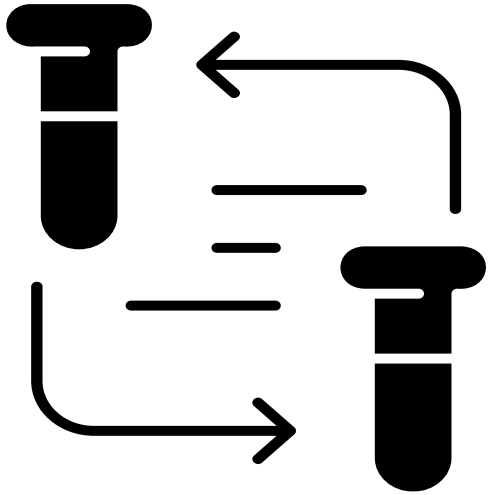


**Scalable**

Many samples

↓

Parallel processing

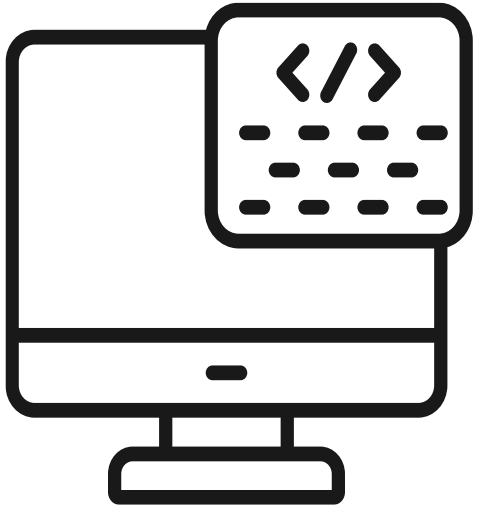


**Reproducible**

Defined steps

↓

Same results



**Shareable**

Reusable across teams

# When to use a workflow?

| Make a Workflow When… | DON’T Make One When… |
|---|---|
| It’s **repetitive** | You’re **still figuring things out** |
| It’s **multi-step** | It’s **exploratory** |
| It’s **multi-sample** | It’s **tiny** |
| It needs to **scale** | It’s **single-step** |
| It needs to be **reproducible** | It’s **temporary** |

# So how do you write a workflow?

There are special workflow languages (code) in which you can give instructions… we like

**Readable**

*workflows look like step-by-step instructions*

**Modular**

*tools are separated into reusable tasks*

**Built for genomics pipelines**

*designed for large bioinformatics analyses*

# How do I get started with WDL workflows?

## You will usually need 3 files

### Ingredients

*What data to use*

**`inputs.json`**

Specifies **inputs**

File paths (FASTQs, BAMs, etc.)

Defines any input values for a task

### Equipment

*What resources to use*

**`options.json`**

Specifies memory to use

What compute to use (HPC, Cloud, etc.)

Specifies where outputs go

### The Recipe

*What tasks to run*

**`workflow.wdl`**

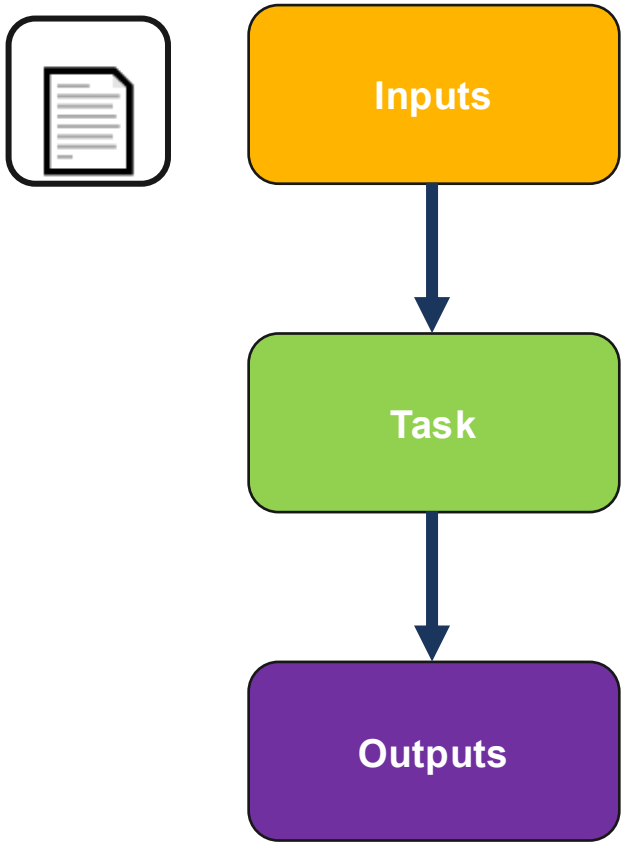
Defines the tasks

Connects inputs → tasks → outputs

Declares dependencies between steps

# hello_world **Workflow**

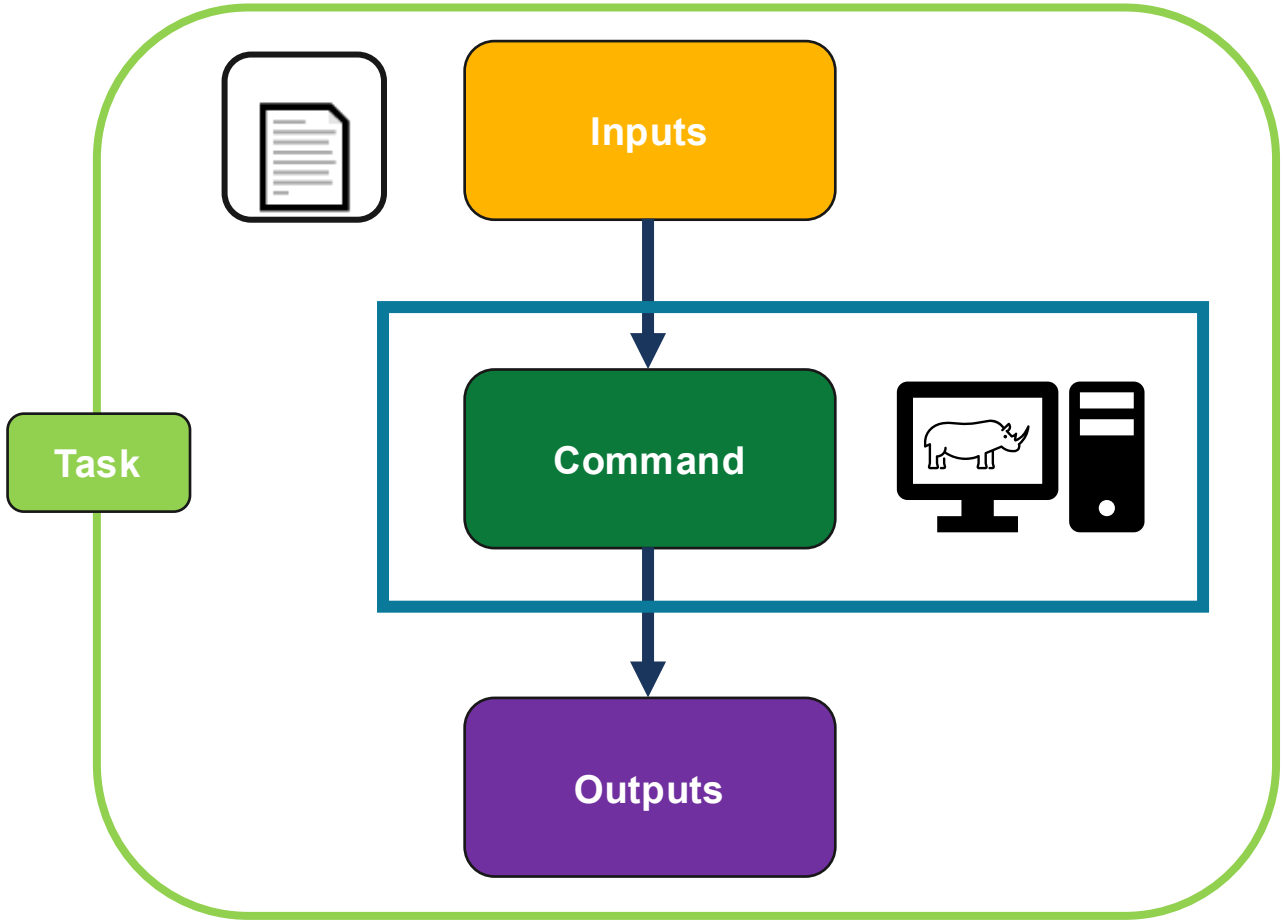
## Human-readable overview of what's happening




```
workflow hello_world {
  meta {
    description: "Writes a personalized greeting"
    outputs: {
      out_file: "A file containing a greeting"
    }
  }

  parameter_meta {
    your_name: "Name to greet"
    your_number: "Your favorite number"
  }

  input {
    String your_name = "Fred"
    Int your_number = 29
  }

  call greeting {
    input:
      name = your_name,
      number = your_number
  }

  output {
    File out_file = greeting.out
  }
}
```

# “Hello World” Task: greeting

Very similar vibes, but at the “building-block” task level:

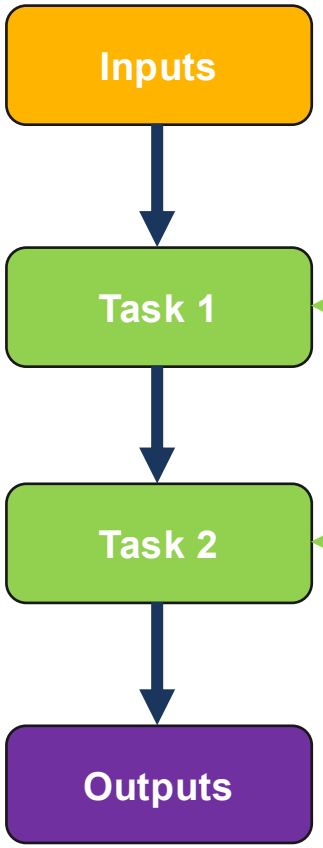



```
task greeting {
  meta {…
  }

  parameter_meta {…
  }

  input {
    String name
    Int number
  }

  command <<<
    echo "Hello, ~{name}!" > "~{name}.txt"
    echo "Your favorite number is ~{number}" \
      >> "~{name}.txt"
  >>>

  output {
    File out = "~{name}.txt"
  }

  runtime {
    docker: "getwilds/awscli:2.27.49"
    memory: "2 GB"
    cpu: 1
  }
}
```

# A real-world example: ww-sra-salmon

## Benefits of WDL become clear at scale…

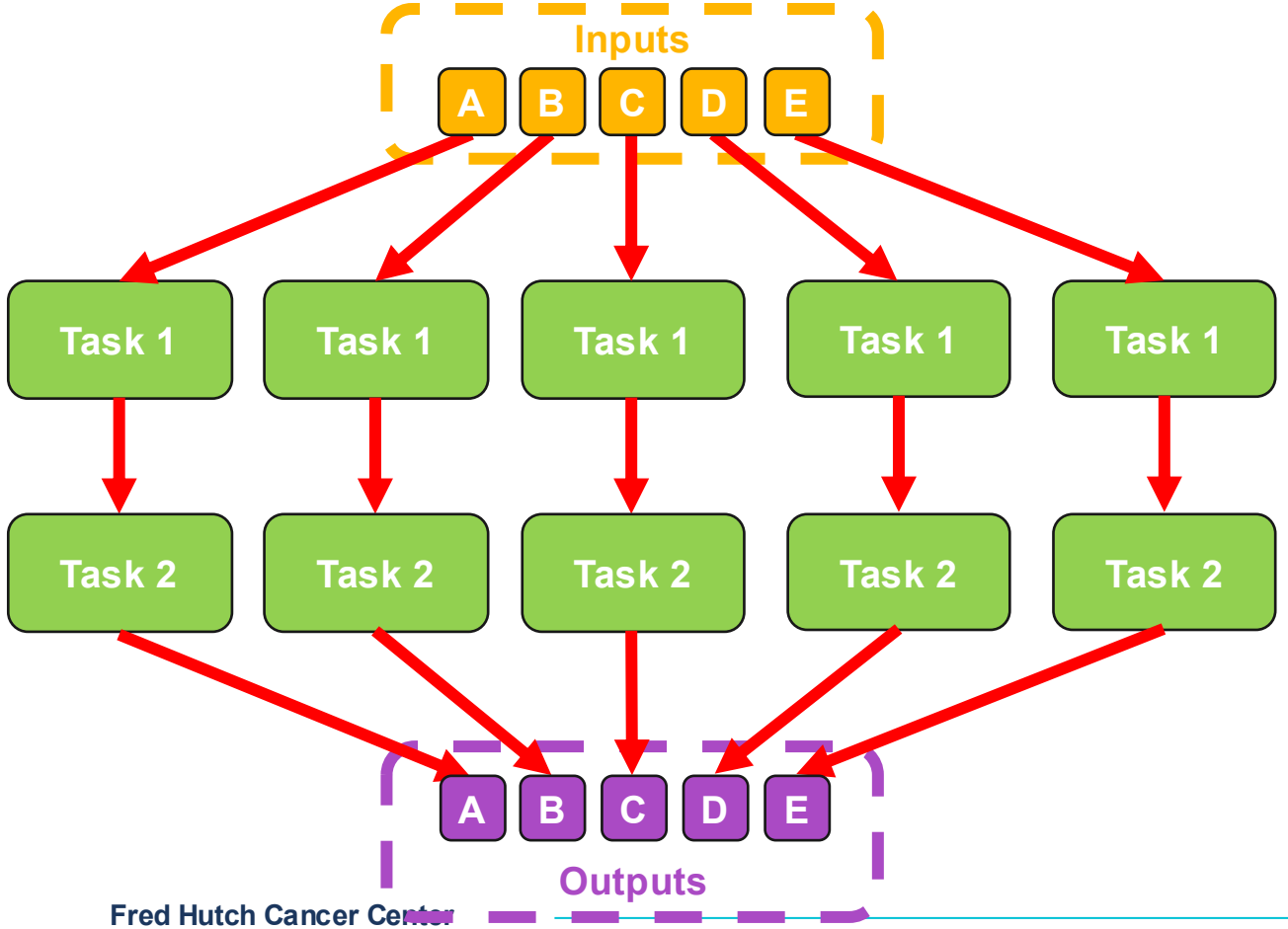
1. Specify what data I'm interested in
2. Download data from source
3. Post-process as necessary
4. Save final results locally




```
import modules/ww-sra/ww-sra.wdl as sra_tasks
import modules/ww-salmon/ww-salmon.wdl as salmon_tasks

workflow sra_salmon {
  input {
    Array[String] sra_id_list
    File salmon_index_zip
  }

  # Download FASTQ files from SRA
  call sra_tasks.fastqdump { input: sra_id = id }

  # Quantify each sample with Salmon
  call salmon_tasks.quantify { input:
      salmon_index_dir = salmon_index_zip,
      fastq_r1 = fastqdump.r1_end,
      fastq_r2 = fastqdump.r2_end,
  }

  output {
    Array[File] quant_dirs = quantify.salmon_quant_dir
  }
}
```

# A real-world example: ww-sra-salmon

Benefits of WDL become clear at scale…




```
import modules/ww-sra/ww-sra.wdl as sra_tasks
import modules/ww-salmon/ww-salmon.wdl as salmon_tasks

workflow sra_salmon {
  input {
    Array[String] sra_id_list
    File salmon_index_zip
  }

  # Scatter across nodes for max parallelization
  scatter ( id in sra_id_list ){

    # Download FASTQ files from SRA
    call sra_tasks.fastqdump { input: sra_id = id }

    # Quantify each sample with Salmon
    call salmon_tasks.quantify { input:
        salmon_index_dir = salmon_index_zip,
        fastq_r1 = fastqdump.r1_end,
        fastq_r2 = fastqdump.r2_end,
    }
  }

  output {
    Array[File] quant_dirs = quantify.salmon_quant_dir
  }
}
```

# Customize the workflows as needed

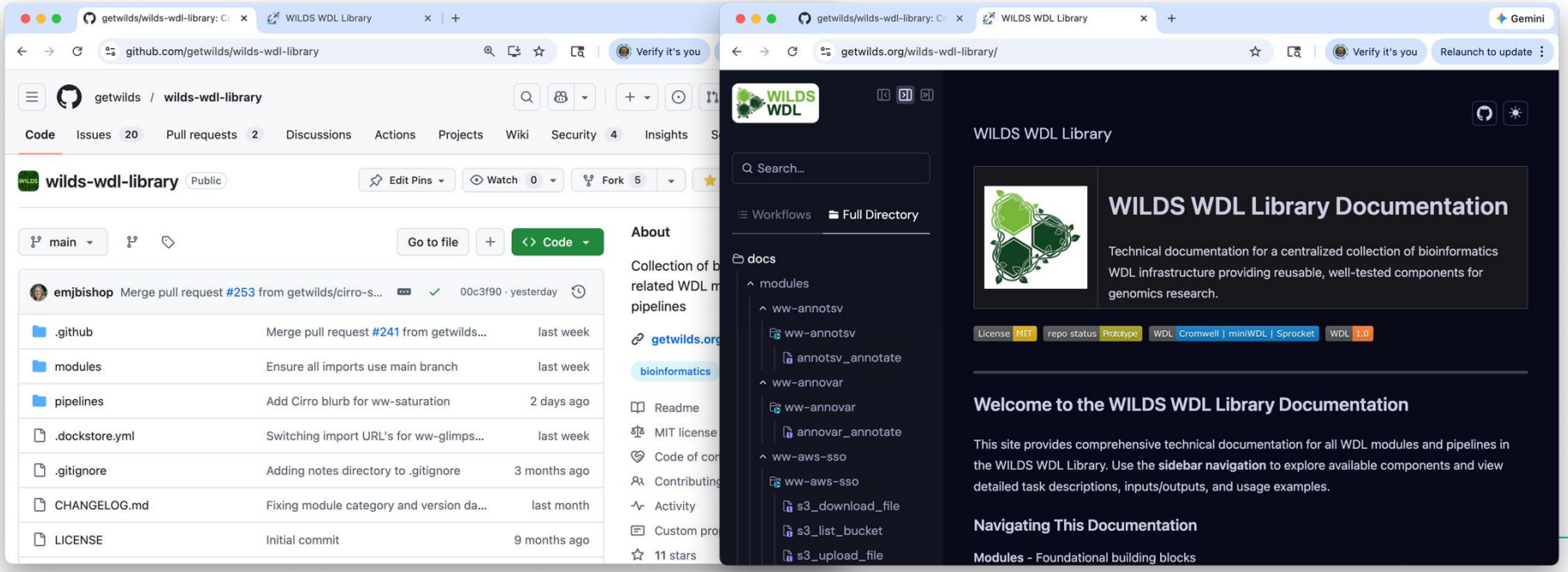
## Straightforward to customize a workflow to suit your research needs

| Problem | Solution | Bioinformatics Example |
|---|---|---|
| I want to provide my own datasets as input… | Update the inputs to accept a list of files! | Array[File] data type<br>[file1.txt, file2.txt, …] |
| I'd prefer to use a different tool for post-processing… | Swap out the import for whichever one you prefer! | ww-salmon → ww-star<br>DNAseq → ww-bwa + ww-gatk |
| I need to add an extra post-processing step… | Import another module and tack a task onto the bottom! | RNAseq → ww-deseq2<br>QC → ww-fastqc |
| I want to save my results to an S3 bucket… | You guessed it… there's a module for that!! | ww-aws-sso<br>(Upload & Download) |
| I don't see a module that fits my needs… | We'll help you build one!!! | WILDS WDL Library<br>WILDS Docker Library |

# Easiest way to write a WDL: use someone else's!

## Plenty of existing WDL workflows out there… including right here at Fred Hutch!

- WILDS WDL Library: curated collection of pre-built, well-tested, well-documented workflows
  - We can help you build one via the WILDS WDL Development Program!!




More details here!!!

**Other Resources**

Broad's WARP
GATK Workflows
BioWDL
Dockstore

# Let’s see workflows in action!

# Demo: hello-world.wdl

FredHutch / data-deep-dives

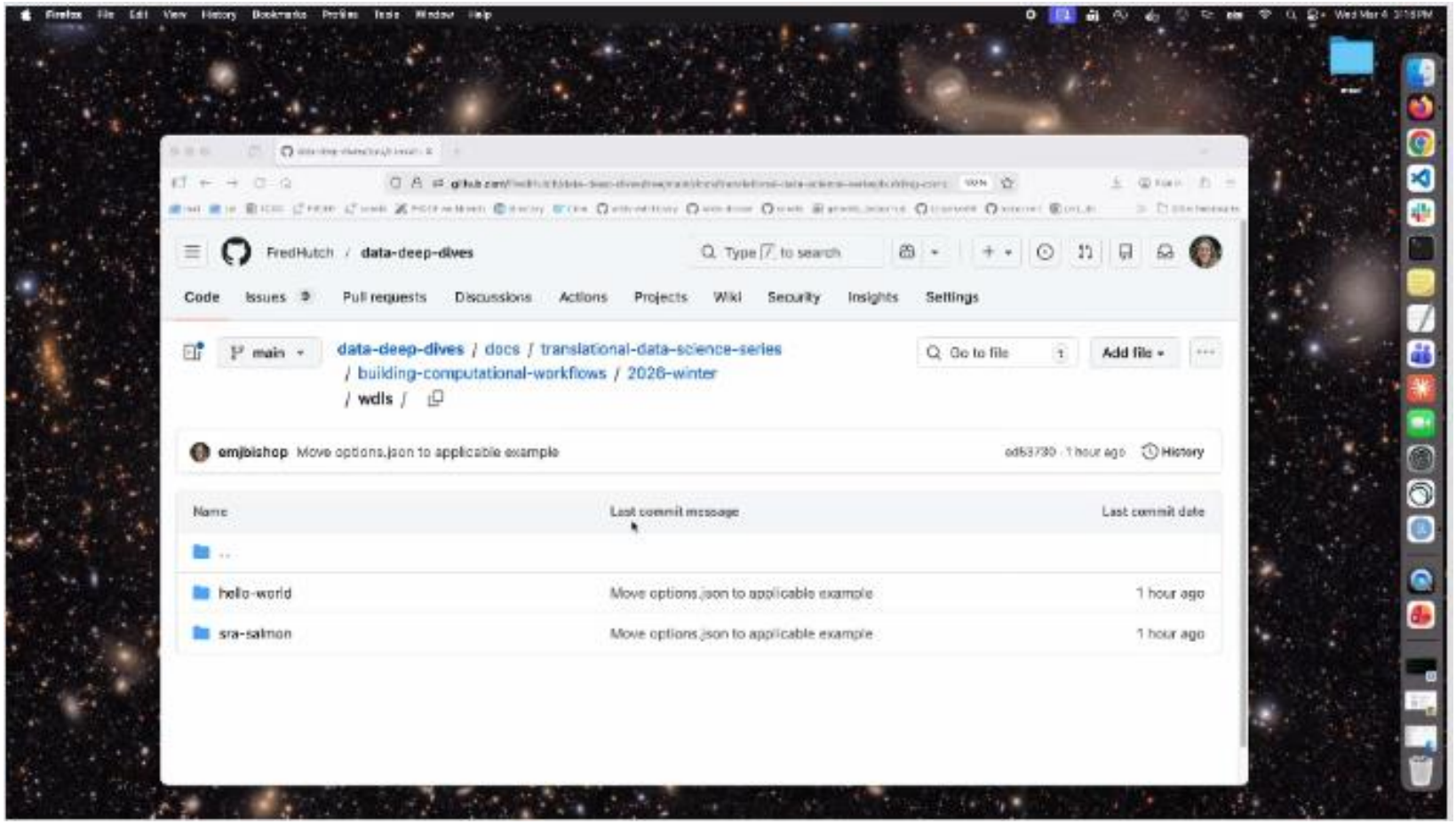
Code | Issues 9 | Pull requests | Discussions | Actions | Projects | Wiki | Security | Insights | Settings

main — data-deep-dives / docs / translational-data-science-series / building-computational-workflows / 2026-winter / wdls / hello-world /

sitapriyamoorthi Rename docs/translational-data-science-series/building-computational-... 2339748 · 4 hours ago History

| Name | Last commit message | Last commit date |
|---|---|---|
| .. | | |
| hello-world-inputs.json | Rename docs/translational-data-science-series/building-computat... | 4 hours ago |
| hello-world.wdl | Rename docs/translational-data-science-series/building-computat... | 4 hours ago |
| options.json | Rename docs/translational-data-science-series/building-computat... | 4 hours ago |

# Demo: sra-salmon.wdl



# Running WDLs at Fred Hutch

| Tool | Interface | Can run | Good for |
|---|---|---|---|
| **Sprocket** | Command line | Any WDL | Development, running workflows manually |
| **PROOF Workbench** | Website | Any WDL | Running at scale, validating before moving to the cloud |
| **Cirro** | Website | Cirro WDLs | Using established pipelines |

**Also popular: miniWDL, Terra, AnVIL**

# Tips for Success

- **Test then scale:** Start with a few samples first
- **Spot-check outputs:** Ensure they look reasonable
- **Check workflow status:** Confirm it started and didn’t fail
- **Read logs to troubleshoot:** Scan for error messages to get context

💬 Reach out on Slack, we've probably seen your issue before

## Fred Hutch WDL Resources

**WILDS WDL Library**
Validated WDLs

**WILDS Docker Library**
Validated Docker images

**PROOF Workbench**
Run WDLs on the cluster through a website

## Fred Hutch Data Science Help

**SciWiki**
Useful knowledgebase

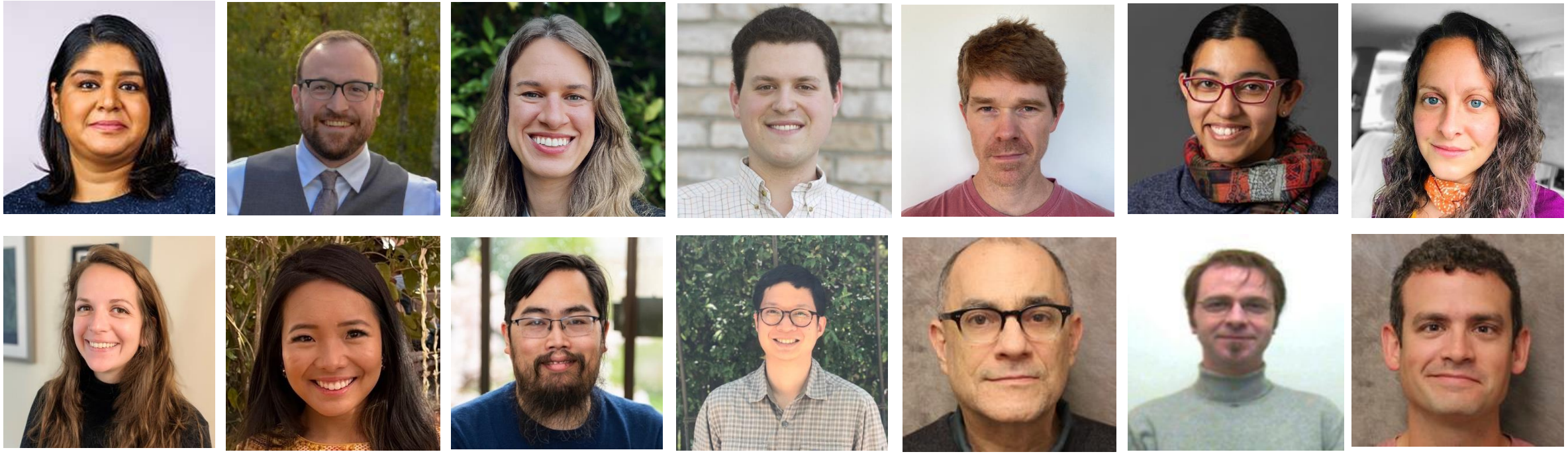
**SciComp**
Get help with the cluster, the cloud, data storage

**FH-Data Slack**
Crowdsourcing, Q&A

**Data House Calls**
Short meetings for personalized help

# Thank you to everyone who made this Deep Dive possible!

And thank you all for participating!!



**Fred Hutch Office of the Chief Data Officer (OCDO)**

**Fred Hutch Scientific Computing (SciComp)**

Fred Hutch
Cancer Center

# Thank you